An energy budget

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect work, motion, heating and an honest model conclusion.

SUCCESS CRITERIA - I can...

1. Keep all terms in consistent energy units.
2. Distinguish mechanical-energy change from total-energy loss.
3. Report assumptions with the energy budget.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “energy budget” and “system” to explain your model or result. Sentence starters are optional.

energy budget: an account of all energy stored and transferred

system: the object or objects chosen for analysis

thermal energy: internal energy associated with microscopic motion/interactions

01 Warm-up: bring back an earlier idea

A 0.3 kg object moving at 5 m/s stops through friction over 0.7 m.
Find energy transformed and average opposing-force magnitude.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $E_i + E_{in} = E_f + E_{out}$.
i/f = initial/final stored energy; in/out = transfers across the chosen system boundary. Account for every form.

Worked reasoning: An input of 100 J gives 60 J useful motion, 25 J thermal transfer and 15 J sound/other transfers. The account balances because 60+25+15=100 J.

Your turn: Draw proportional energy arrows and calculate the useful fraction.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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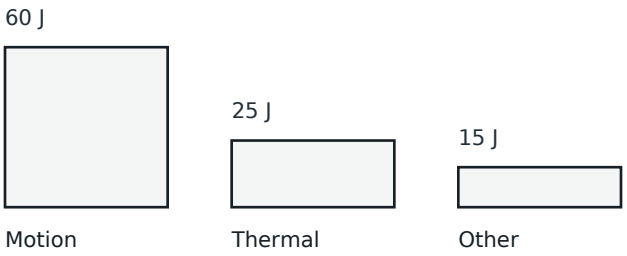
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

A modeled system starts with 100 J. It ends with 60 J motion energy, 25 J thermal energy and 15 J sound/deformation.
Draw an energy account and check the total.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: 1 2 3

04 Apply the physics

A simulated 1000 kg probe has $K = 8 \times 10^6$ J and gravitational $U = -6 \times 10^6$ J, with $U = 0$ at infinity.
Find total mechanical energy. Drag then removes 3×10^6 J. Complete the energy account and discuss whether the final state is bound in an isolated central-gravity model.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

05 Explain your reasoning

What information is missing if an impact report gives only total kinetic energy and says all of it heats the ground?

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

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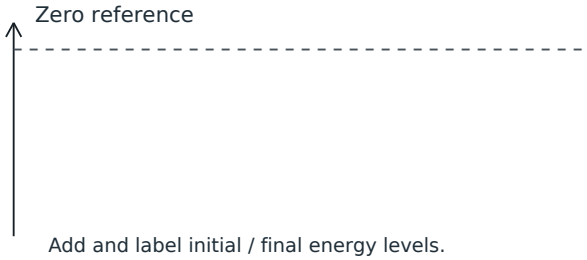
Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

With GMm=192 J·m, move an object from r=4 to r=8 m.
Find work by gravity using U=-GMm/r.
Label the distances or energy levels used in your reasoning.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Add and label initial / final energy levels.

Marzano check - circle: 1 2 3

07 Exit check

A 2 kg cart descends 3 m from rest with g=10. Losses are 12 J.
Find bottom speed.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both ... . They differ in ... because ... .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A device takes 600 W for 5 s and raises a 10 kg load 12 m using g=10.
Find useful work, efficiency and remaining energy.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both ... . They differ in ... because ... .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3